

Green Chemistry

Air contains: 78% nitrogen, 21% oxygen, 1% noble gases

How can we separate gases in air?

- ① Air is cooled under high pressure until it turns to liquid.
- ② Liquid air is separated using fractional distillation, which happens because gases have different boiling temperatures. So when liquid air is boiled, the gases boil at different temperatures, and can be collected one by one.

The order of which gases boil first:

Nitrogen $\rightarrow$ Argon $\rightarrow$ Oxygen $\rightarrow$ Krypton $\rightarrow$ Xenon.

Pollutant	Caused by	Problems caused	Can be reduced by
Carbon dioxide, CO₂	The complete combustion of carbon-containing fuels	Higher levels of carbon dioxide lead to increased global warming, which leads to climate change	Burning fewer fossil fuels
Carbon monoxide, CO	The incomplete combustion of carbon-containing fuels	Toxic gas	Fitting catalytic converters to remove CO, but more CO ₂ is produced. It should be noted that lean-burn engines also produce less CO.
Particulates, C	The incomplete combustion of carbon-containing fuels	Increased risk of respiratory problems and cancer	Adding oxygenates such as ethanol to petrol to reduce the particulates produced
Methane, CH₄	The decomposition of vegetation and waste gases from digestion in animals	Higher levels of methane lead to increased global warming, which leads to climate change	Encouraging composting or incineration. Capturing the methane produced and using it as a fuel or to generate electricity when burned. Reducing the number of farm animals by encouraging people to eat less meat
Nitrogen oxides, NO_x	Car engines / Furnaces / Places with thunderstorms from the energy in lightening	Acid rain, photochemical smog and respiratory problems	Fitting catalytic converters to remove nitrogen oxides. However, in doing so more carbon dioxide is produced.
Sulfur dioxide, SO₂	The combustion of fossil fuels which contain sulfur compounds (coal, oil, gas) Volcanoes (Natural source)	Acid rain	Removing sulfur from petrol to produce low-sulfur petrol. Using flue gas desulfurisation units (see p. 182) at coal-burning power stations

→ Nitrogen monoxide, from the reaction between N₂ & O₂ at high temperature

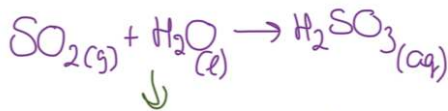
→ Reacts with oxygen after being released from the exhaust to form brown acidic gas: Nitrogen (IV) oxide:

(Nitrogen dioxide)

→ Dissolves in water to produce nitric acid → causes acid rain



dissolves in rain water to form the weak acid, Sulfurous acid:



Sulfurous acid is then further oxidized to form Sulfuric acid

↓
Causes acid rain

Effect of acid rain:

- 1- leaves will be damaged by the acid, falling off the tree and dying
- 2- lakes and rivers become acidic, which may kill the aquatic life
- 3- Soil will become too acidic to grow crops. And it causes the minerals to dissolve
- 4- Buildings made of carbonate rocks will be eroded
- 5- Metal structures such as bridges and iron railings will corrode

To solve acid rain problems:

1- Use low sulfur fuels: Coal and petroleum are treated to remove sulfur compounds.

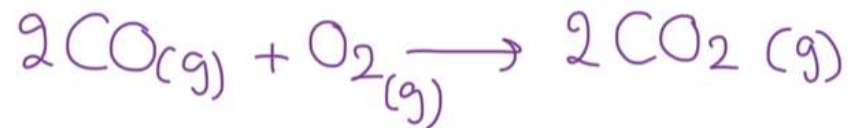
2- Flue gas desulfurization: Removing sulfur dioxide from flue gases by treatment with calcium oxide (CaO: lime), which removes sulfur dioxide by producing calcium sulfate.

3- Exhausts of cars are fitted with catalytic converter:

1- Nitrogen monoxide is reduced to nitrogen and oxygen



2- Hydrocarbons & carbon monoxide are oxidised.



a catalyst is used to speed up reactions:

- * Platinum
- * Palladium
- * Rhodium

Greenhouse gases: CO_2 and CH_4 : Absorb heat from the atmosphere and prevents it from escaping to space. They cause global warming.

Pollutant	Caused by	Problems caused	Can be reduced by
Carbon dioxide, CO_2	The complete combustion of carbon-containing fuels	Higher levels of carbon dioxide lead to increased global warming, which leads to climate change	Burning fewer fossil fuels
Carbon monoxide, CO	The incomplete combustion of carbon-containing fuels	Toxic gas	Fitting catalytic converters to remove CO , but more CO_2 is produced. It should be noted that lean-burn engines also produce less CO .
Particulates, C	The incomplete combustion of carbon-containing fuels	Increased risk of respiratory problems and cancer	Adding oxygenates such as ethanol to petrol to reduce the particulates produced
Methane, CH_4	The decomposition of vegetation and waste gases from digestion in animals	Higher levels of methane lead to increased global warming, which leads to climate change	Encouraging composting or incineration. Capturing the methane produced and using it as a fuel or to generate electricity when burned. Reducing the number of farm animals by encouraging people to eat less meat
Nitrogen oxides, NO_x	Car engines	Acid rain, photochemical smog and respiratory problems	Fitting catalytic converters to remove nitrogen oxides. However, in doing so more carbon dioxide is produced.
Sulfur dioxide, SO_2	The combustion of fossil fuels which contain sulfur compounds	Acid rain	Removing sulfur from petrol to produce low-sulfur petrol. Using flue gas desulfurisation units (see p. 182) at coal-burning power stations

→ Caused by: 1- The complete combustion of carbon compounds



2- Deforestation and burning of trees and vegetation.

3- Respiration: The reaction between glucose and oxygen



4- The reaction between dilute acids and carbonates



→ Forms when bacteria breaks down plant material or carbon-based material in the absence of oxygen. In paddy fields, swamps, landfill sites.

* Grazing animals, as they give out methane as waste gas, when bacteria in their stomachs break down grass and other food

Effect of climate change:

- Dry places will become drier. Frequent drought in other wetter places.
- Melting land-ice in the Arctic & Antarctica, which will cause sea levels to rise, so low-lying countries will be at risk of flooding
- Glaciers in high mountain ranges will shrink away, reducing the annual supply of meltwater for rivers
- Storms, floods, and wildfires are more severe and frequent
- Farmers won't be able to grow their usual crops
- Species that cannot adapt to changing climate will become extinct
- Harmful insects and diseases will spread to new countries that have grown warm enough for them.

Tackling climate change:

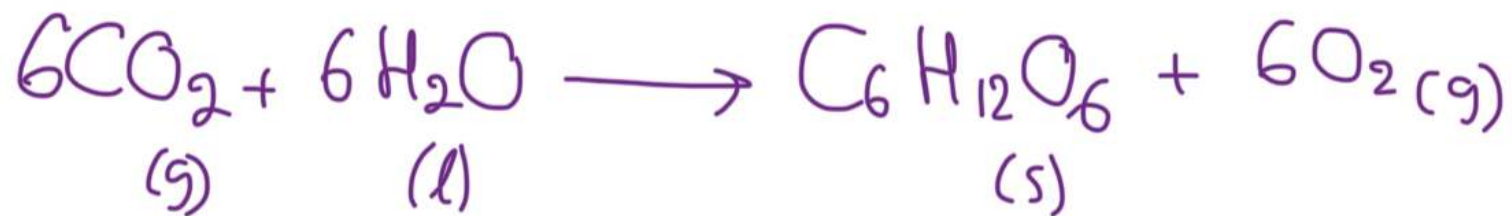
1- Using non-carbon fuels:

- 1- Wind, waves, tides, and fast flowing rivers
- 2- Nuclear power
- 3- Solar power

2- Non-carbon power for transport:

- 1- Hydrogen - oxygen fuel cells
- 2- batteries that are charged using non-carbon electricity

3- By planting trees to get rid of Carbon dioxide by Photosynthesis



4- By reducing methane emissions:

- * Farmers reduce growing animals
- * Change human eating habits, reducing meat and dairy products

Uses of water :

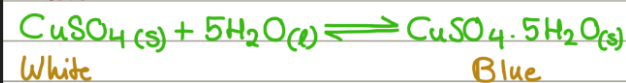
- 1- At home: * Cooking
* Cleaning
* Drinking
- 2- In industry: * As a solvent
* As a coolant
* For cleaning
* As a chemical reactant.
- 3- In power stations: * heated to make steam. the steam drives the turbines that generate electricity.
- 4- In Farms: * Drink for animals
* Watering crops.

Tests for water:

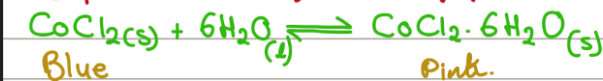
* Physical test: To test for the purity of water :- Boils at 100°C
- Freezes at 0°C

* Chemical test: To test for the presence of water (pure or impure):

1- Add the unknown liquid to anhydrous copper(II) sulfate. If it changes from white to blue then the solution is water.



2- Dip blue cobalt(II) chloride paper into the liquid. It'll change to pink if the liquid is water.



* Distilled water is used in practical chemistry rather than tap water because it contains fewer chemical impurities.

Water from natural resources contain:

1- Beneficial substances:

* Dissolved oxygen for aquatic life

* Some metal compounds that provide essential minerals:

- Calcium for bones and teeth.
- Magnesium and calcium help protect against heart disease.
- Sodium, potassium, iron, copper, zinc, Fluorine

2- Harmful substances:

* Toxic metal compounds:

- Lead causes brain damage
 - Mercury damages the nervous system
 - Arsenic increases the rate of cancer and heart disease.
- } → Drain in from mines and factories
- } → Natural in rocks

* Plastics: The aquatic animals mistake it for food. They cannot digest it, and it clogs their guts → Starving.

* Microbes from untreated sewage: Include bacteria that cause Cholera, Typhoid, and the virus hepatitis.

* Nitrates and phosphates from agricultural waste and detergents: They lead to deoxygenation of water and damage to aquatic life.

The treatment of domestic water:

- 1- Impure water is passed through screens; a screen traps bigger solids such as plastic bottles and twigs.
- 2- Filtration through coarse sand traps larger, insoluble particles.
- 3- A sedimentation tank has flocculants, aluminum sulfate. They make fine, suspended particles clump together, so that they form larger clumps, and sink to the bottom of the tank. Some toxic metal compounds will stick to the clumps.
- 4- These particles are removed by further filtration through fine sand.
- 5- Carbon (charcoal) is used to remove unwanted tastes and odours.
- 6- Little chlorine gas is added, which sterilises the water and kills microbes. (Chlorination)

→ This process removes the tiniest particles. Steps 2 & 5 remove toxic metals. Chlorination kills all microbes. But the water still contains harmful substances dissolved in it, like high levels of nitrates from fertilisers, which can be dangerous

* Fluorine is sometimes added to prevent tooth decay.